Soft Computing for Visual Terrain Perception and Traversability Assessment by Planetary Robotic Systems

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Abstract - *This paper discusses technical challenges and navigational skill requirements of mobile robots for traversable path planning in natural environments similar to Mars surface terrains. Different methods for detecting salient terrain features based on imaging texture analysis techniques are described. In particular, three competing soft computing techniques are presented for terrain traversability assessment: a rule-based terrain classifier, a neural network-based terrain classifier, and a fuzzy-logic terrain classifier. Each terrain classifier divides a region of natural terrain into finite sub-terrain regions and classifies terrain condition exclusively within each sub-terrain region based on terrain visual clues. Image processing techniques are applied for aggregative fusion of sub-terrain assessment results. Results of a comparative performance evaluation of all three terrain classifiers are presented. The last two terrain classifiers are shown to have remarkable capability for traversability assessment, which facilitates navigation in unstructured natural terrain environments.*

Keywords: Terrain Traversability Assessment Modeling, Visual Terrain Mapping, Image Texture Analysis

1. Introduction

Autonomous navigation in natural terrain is an emerging technology of interest to the U.S. Department of Defense for military surveillance, reconnaissance, and target tracking applications and, in particular, to NASA for robotic planetary space exploration. To operate autonomously and intelligently in unstructured terrain, a robot must possess many advanced navigational skills. In particular, it needs to have capability to perceive its surrounding environment, recognize hazardous terrain topography, discriminate negotiable and non-negotiable obstacles, and select suitable routes compatible with its mobility capabilities without jeopardizing its safety and mission.

A robot's endurance for long-range navigation in natural terrain environments is dependent upon its capability to perceive its environment reliably and adapt to it vigorously. Without terrain adaptability, the control scheme of the robot will experience many difficulties when attempting to navigate the robot safely. Terrain adaptability requires calculated terrain assessment and situation-awareness. Present robotic technology has begun to address some of these shortcomings [1]. Future robotic military and space applications, however, demand more intelligent robots with robust terrain adaptability.

In this paper, we have addressed issues related to visual terrain sensing, mapping and path planning of mobile robots. In section 2, we have presented some background on the related research works in this area. In section 3, we have discussed navigational skill requirements of intelligent mobile robots. In section 4, we have discussed details of three terrain classifiers for terrain traversability assessment and visual terrain mapping purposes. In section 5, we have presented technical discussions on performance of the terrain classifiers. In section 6, we have compared performance of the three terrain classifiers. Finally, in section 7, we have presented the conclusion of this paper.

2. Background

Robot navigation in natural terrain is challenging due to uncertainty in recognition of salient features of the environment. Natural terrain is deceiving, unpredictable, and intricate. Without a good perceptive model of the terrain, a robot cannot reliably navigate autonomously to accomplish its mission successfully. For example, the NASA *Spirit* and *Opportunity* rovers successfully navigate on Mars by performing onboard traversability assessment and local path planning using the GESTALT algorithm [2]. Geometric analysis of nearby terrain is performed using dense range data to generate a local traversability map of step, slope, and roughness hazards through which to plan safe paths. Terrain traversability and path planning of terrain-adaptive robots have been addressed by a number of researchers. Howard et. al., [3,4] presented a technique for terrain traversability assessment learning for outdoor navigating mobile robots. Using human-embedded logic in real-time, they

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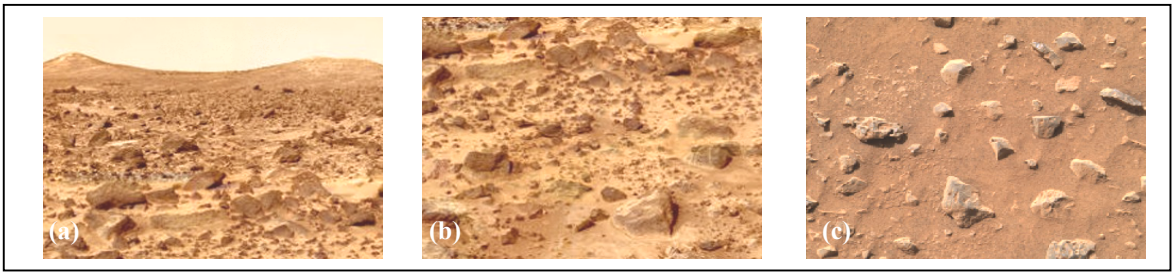


Figure 1. Three Navigational Planning Situations in Natural Terrain (images courtesy NASA/JPL-Caltech): (a) Long-range Route Planning, (b) Intermediate-range Navigational Planning, (c) Short-range Mobility Planning.

demonstrated a technique for development of terrain perception based on features extracted from imagery data. In their method, they introduced a fuzzy logic framework and vision algorithms for analysis of terrain. The work herein evaluates three different techniques for terrain classification other than those proposed in [3, 4], including an alternative neural network architecture. Golda et. al., [5] presented a probabilistic modeling technique suitable for analysis of high-speed rough-terrain mobile robots. They have experimentally shown that their model can accurately predict robot performance in simple, well-known terrain, however, in an unstructured environment, the performance of their stochastic model degraded. A combination of terrain complexity and unaccountable uncertainty measures were found as leading causes in degradation of their predictive terrain assessment model.

Using stereo vision-based terrain mapping, vast knowledge about the terrain environment can be obtained rather rapidly to generate a reference model or map used to represent pertinent terrain landmarks as well as for long-range traverse planning purposes. Without such maps, a robot may spend much of its time and energy venturing along what turns out to be a dead end. Olson et. al. [6] proposed a method based on visual terrain mapping for Mars rovers. Using a visual stereo imaging fusion technique, they have demonstrated a reliable method for high fidelity terrain mapping and robot world perception modeling. By compiling terrain map images using a system that unifies multi-resolution models, they were able to integrate Mars rover, lander descent, and orbital images to obtain 3D terrain maps used for rover traverse planning.

Another notable challenge is that of localization of the robot and terrain mapping as it traverses the environment. Many researchers have addressed this problem in the context of well-structured (indoor) environments and have obtained important results [7-9]. These algorithms have been implemented using different sensory modalities, such as camera vision systems [10], laser range sensors [11], and ultrasonic sensors [12]. In these techniques, the robot is usually moved sequentially, by either following topological graphs, or using *greedy* algorithms that explore regions only on the extreme edges of the known environment [10,11,13]. The environment is typically modeled using geometrical descriptions in several ways, including graph models and Voronoi diagrams, occupancy grid models, segment models, vertex models, and convex polygon models [12,13]. The focus of these works is accurate mapping of the structured environment. They do not address mapping efficiency and uncertainty issues related to natural terrains. They generally assume that the environment is effectively flat and readily traversable [10-14].

2. Natural Terrain Navigation

The robot adaptability to its environment is a crucial key for navigational endurance and longevity of the robot in natural terrain. To be adaptable, a robot must perceive its environment and reason about circumstances surrounding it. Autonomous robot navigation in natural terrain can be divided into three hierarchical related tasks: (1) *Route Planning*, (2) *Navigation Planning*, and (3) *Mobility Planning*. As illustrated in Figure 1, in *Route Planning,* the objective is that of assessment of the terrain in bulk from far away. This may include determining feasible routes that maximize traversability potential of the robot as well as its safety and health. Another objective is to identify pertinent intermediate landmarks, i.e., attractive features of the environment that the robot can easily identify without ambiguity. Terrain landmarks can be used for localization of the robot. The landmarks are also waypoints that the robot can easily follow along its path and measure its movement progress relative to. A typical route plan may consist of many waypoints laid out on either an aerial map or a landscape image captured from an elevated ground, or a series of global position coordinates defined manually. A typical range of applicability for route planning is from tens to thousands of feet depending on visibility and intricacy of the terrain.

Navigation planning considers the locations of obstacles, treacherous hills and slopes, and positive and negative terrain elevation features in short range. A set of short traverse path segments from current robot location to next intermediate waypoint or the goal is determined. Localization and navigation error recovery are critical at this phase, in order to keep the robot as close as possible to its designated path or model frame of reference between its intermediate landmarks. Due to loss of environmental details in the field of view, navigation certainty in correct recognition of landmarks tends to

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diminish with the distance of landmarks from the robot. To manage the navigation uncertainty, typically the range of effectiveness of localization is within a few feet radius from the robot. Depending on the navigational speed of the robot, terrain intricacy, and sensory reliability, this range can be adjusted appropriately. Typical range of applicability of navigational planning varies from 2 to 12 feet for a slow moving robot and to a higher range for a fast moving robot.

The purpose of *Mobility planning* is to formulate a set of low-level drive actions that moves the robot through the terrain while negotiating and/or avoiding obstacles along its path. An important consideration in this phase is the dynamic interaction of the robot with its environment, in particular, detection of robot wheel traction losses during driving and steering while avoiding wheel traps, wheel supports, and tip over states [15]. Another consideration is optimization of the robot's safety, energy, and reliability [5]. Typical range of applicability of mobility planning is from a few inches to a few feet depending on terrain complexity, physical size of the robot's wheels, and the robot's body clearance from the ground. In [15], dynamic modeling of robot and environment is shown to be reliable for simple, well-characterized terrain. However, on rough or unknown terrain, model fidelity degrades due to imprecise knowledge of terrain parameters.

As explained above, it is clear that an autonomous terrain-adaptive robot requires different tactical skills in order to able to navigate an unstructured terrain. Both long and short range path planning are necessary to sustain the traversability longevity of the robot. In the following discussion, we discuss methods for terrain traversability assessment for terrain-adaptive robots.

4. Terrain Traversability Assessment

As a robot traverses natural terrain, images may be periodically acquired for traversability assessment. This assessment is crucial for all three navigational tasks described earlier. Close proximity of robot cameras to the ground plane permits very useful details about salient features of the terrain to be detected at this stage and characterized with a high degree of confidence. At each navigation phase, condition of the terrain can be classified based on the relevant features extracted from the images. Objects in different natural terrain have certain distinctive visual texture properties. The common surface texture attributes include: contrast, variance, energy, entropy, and homogeneity. Image texture is, basically, due to geometrical arrangement or occurrence of certain pixel intensity variations in repeated pattern(s). Repeated patterns in images are caused by illumination of terrain object surfaces, such as scattered rock formations on the many rocky terrains of Mars. Rock surface texture characteristics on Mars vary considerably in roughness, waviness, flaws, cavities, uneven surface pits, and colors.

One reliable way to perform terrain traversability assessment is to apply image texture quantitative statistics in conjunction with qualitative soft computing techniques. Three soft computing techniques for terrain traversability assessment are presented below: (1) heuristic rule-based system, (2) neural network, and (3) fuzzy logic reasoning. First, we describe our underlying method for terrain texture characterization.

4.1 Terrain Texture Characterization

An image is a matrix of pixel intensities, $I_{i,j}$ bounded within a certain range. Typically, this range varies from 0 to 255 and corresponds to 256 gray levels. Variation in image pixel intensities creates certain repeated patterns called *Image Texture*. Imaged terrain texture patterns can be used as a means to assess observed conditions of a terrain. Natural terrain has many texture patterns that are difficult to define mathematically. A practical way to classify the condition of a natural terrain is to apply statistical analysis. Terrain texture features can be characterized by statistical means using first, second, and higher order statistics. Since human perception has been found to be sensitive to second order statistics, most recent works in this area are based on the features extracted from the second order gray level such as co-occurrence matrix [16]. We follow a similar approach as described below.

Let $p_s(i,j)$ and $p_d(i,j)$ be the first order difference and summation in intensities of a pair of image pixels i and j with a certain orientation that are d pixels apart. Since image texture can have any size and orientation, the orientation and distance between image pixel pairs is trivial. We also define normalized occurrence summation vector P_s and difference vector P_d such that:

$$P_s = \frac{\hbar[p_s(i,j)]}{2\square} \qquad and \quad P_d = \frac{\hbar[p_d(i,j)]}{2\square} \tag{1}$$

where $\hbar[x]$ represents the histogram of the variable x, $/\!/$ represents the maximum range of shade variation of image pixels, and i and j indicate coordinates of the image pixel in the image coordinate frame. The size of each vector P_s and P_d is $2\square+1$. $P_s(k)$ and $P_d(k)$ represent probabilities of occurrence of image texture pair k. Based on these notations, we define the Image Texture Variance (V) as:

$$V = \sum_{i=0}^{2\square} (i \square b)^2 . P_s(i) + \sum_{i=0}^{2\square} (i \square\square)^2 . P_d(i) \tag{2}$$

where $b = Max(P_s)$. We also define normalized image texture Contrast (C) as:

$$C = \frac{1}{2\square} \sum_{i=\square\square}^{\square} i^2 . P_d(i+\square) \tag{3}$$

The image texture contrast measures the amount of local pixels intensity variation within an image. One other important texture property is *Image Texture Energy (E)* that is a measure of its textural uniformity of the image and we defined it as:

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$$E = \frac{1}{2^{n}}\left[\sum_{i=0}^{2^{n}} P_s^2(i) + \sum_{i=0}^{2^{n}} P_d^2(i)\right] \quad (4)$$

The texture information content of an image is best defined by *Image Texture Entropy* (En), which we define as:

$$En = \sum_{i=0}^{2^{n}} P_s(i).\log(P_s(i)) + \sum_{i=0}^{2^{n}} P_d(i).\log(P_d(i)) \quad (5)$$

The expression above holds for any value of $P_s(i)$ and $P_d(i)$ greater than zero. In order to minimize the computation requirement, we chose the contrast, variance, and energy texture attributes as the basis for our terrain surface texture analysis. We chose this set, in particular, since those attributes provide reliable statistical assessment of terrain surface textures and signify terrain features well under a variety of environmental conditions. For smooth terrain, image texture energy, entropy, and homogeneity measures are high and image texture contrast is low. For a rocky terrain, the image texture energy, entropy, and homogeneity measures are low and image texture contrast measures high. Disparity in terrain texture patterns can be used to consistently classify condition of a terrain and determine a terrain traversability measure based on which a robot can perceive its environment and react to it appropriately. In the following discussions, we present three competing methods for terrain traversability classification and assessment.

4.2 Heuristic Rule-based Terrain Classifier

In our visual terrain approach, we initially divide a terrain image into a matrix of finite sub-frames. Each finite sub-frame represents a small portion of the actual terrain in the image that we call "sub-terrain" (as opposed to "subterranean"). Next, we compute surface texture properties for each image sub-frame and form a Terrain Feature Matrix, M_{fv}, where each element of this matrix represents a sub-terrain feature vector. To obtain an aggregated assessment of the terrain, we apply a 2D *Kalman Filter* to M_{fv} to fuse terrain traversability measures. This process is analogous to an image convolution operation and achieved in two steps. First the *Kalman Filter* is applied row by row to the matrix M_{fv} and a second time it is applied column by column to rectify the matrix M_{fv}. This operation preserves permanence of localized terrain salient features within neighboring sub-terrain areas. This method offers two advantages. First, it allows a much simpler, yet more concise rule-based system to be developed for terrain traversability assessment purpose. Secondly, it offers an opportunity for performing parallel processing since each sub-terrain image can be independently analyzed. This feature can considerably expedite terrain traversability assessment if the robot has onboard computational parallel processing capability.

Our heuristic Rule-Based Terrain Classifier (RBTC) is a knowledge driven deterministic model. It has a set of pre-defined thresholds for classification of different sub-terrain conditions. These threshold settings were derived after examining hundreds sub-terrain condition varieties; in particular, the sub-terrains that contained distinctive salient terrain landmarks. We have considered both partially and fully buried rocks, pebbles and different types of rock formations, sand piles, and even rovers' wheels footprints on previously undisturbed land surfaces of Mars. Table 1 presents the set of crisp rules we derived for classification of Martian sub-terrains. The RBTC mainly utilizes two sub-terrain texture properties, texture contrast and texture entropy. Both image texture properties are measured at 45 degrees. Figures 2 and 3 present results of traversability assessment of different Martian terrains based on this governing heuristic rule-

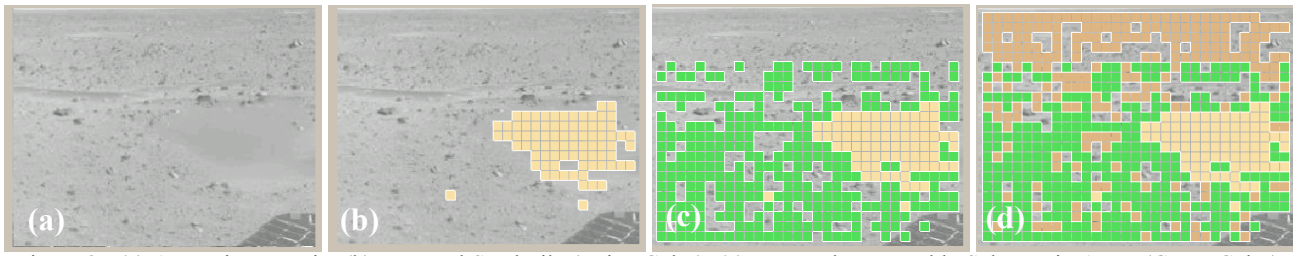


Figure 2. (a) A Martian Terrain, (b) Detected Sand Pile (Beige Color), (c) Detected Traversable Sub-terrain Areas (Green Color) Around the Sand Pile, (d) Terrain Small Rocky Area (Brown Color). Unmarked Areas in (d) correspond to those areas that the algorithm has recognized restricted obstacle formation. They are not color marked intentionally for a better clarity of identified terrain obstacle formations.

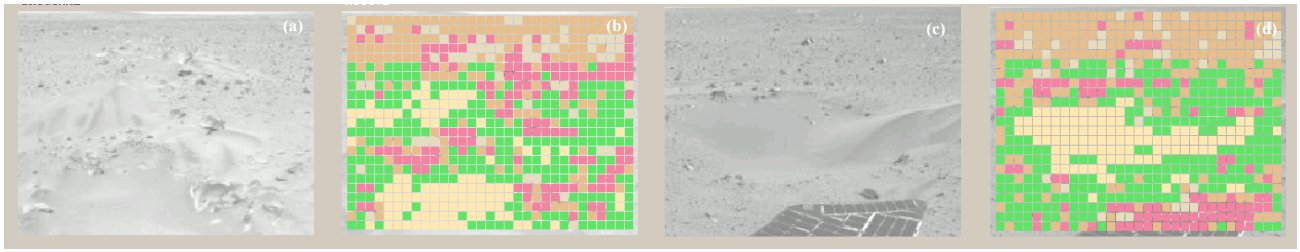


Figure 3. (a) A Martian Terrain with Sand piles and rock formation, (b) Terrain assessment results of image (a); (c) Another image of a Martian Terrain, (d) Terrain assessment results of image (c). Color designations are: Sandy sub-terrains (dark beige color), rocky sub-terrains (brown color), rough sub-terrains (light beige color), impassable sub-terrains (red color).

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based terrain classifier. (These and subsequent terrain images, except for Figure 9, were captured by the *Spirit* rover at Gusev Crater on Mars during the NASA Mars Exploration Rover mission [17], courtesy NASA/JPL-Caltech.)

Table 1. Heuristic rule-base for sub-terrain classification

TERRAIN TEXTURE MEASURES	TERRAIN CONDITION
If (C > 0.01 Or En > 20.0)	Sand Piles
If (C >= 0.01 And C < 0.1) Or If (En>1.50 And En<=25.0)	Smooth Terrain
If (C >= 0.10 And C < 0.2) Or If (En<=1.5 And En > 0.60)	Rough Terrain
If (C >= 0.20 And C < 0.35) Or If (En<=0.6 And En > 0.1)	Rocky Terrain
If (C >=0.35 And C < 0.80) Or If (En<=0.10)	Very Rocky Terrain
If (C >=0.80)	Large Rocks

Where C: Contrast and En: Entropy

4.3 Neural Network-based Terrain Classifier

Our neural network-based terrain classifier (NNTC) is based on a supervised Hamming network that is trained with texture characteristics of sub-terrain as described in section 4.1. The Hamming network is an extension of the Hopfield network in that it adds a maximum likelihood classifier to the front end. Richard Lippman originally developed this network in the mid 1980's [18]. The Hamming network implements a classifier based upon least error for binary input vectors, where the Hamming distance defines the error. The Hamming distance is defined as the number of bits that differ between two corresponding, fixed-length input vectors. The Hamming network has three layers. There is an example network shown in Figure 4. The network uses an input layer with as many nodes as there are separate binary features. It has a category layer, which is the Hopfield layer, with as many nodes as there are categories, or classes. This differs significantly from the formal Hopfield architecture, which has as many nodes in the middle layer as there are input nodes. And finally, there is an output layer, which matches the number of nodes in the category layer. The learning of a Hamming network is similar to the Hopfield methodology in that it requires a single-pass training set. However, in this supervised paradigm, the desired training pattern is impressed upon the input layer while the desired class to which the vector belongs is impressed upon the output layer. Here the output contains only the category output to which the input vector belongs. Again, the recursive nature of the Hopfield layer provides a means of adjusting all connection weights. We applied a revised version of Hamming network that initially converts the real value sub-terrain texture properties to binary input vectors. The sub-terrain texture feature vector that we chose contained Energy, Contrast, Variance, and Image Blob size. The latter property is achieved by measuring the size of largest blob after performing an adaptive binarization operation over a sub-terrain image window area. For training of the network, we

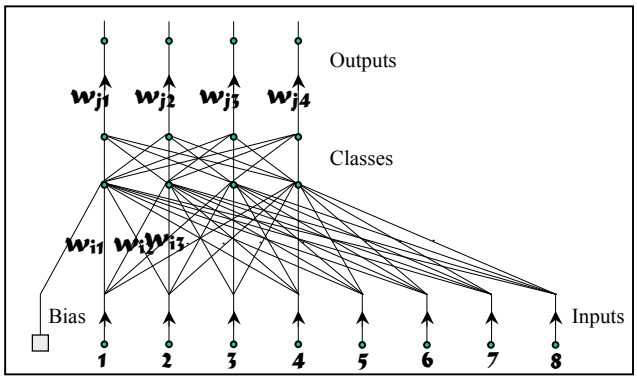


Figure 4. Architecture of the Hamming NN Terrain Classifier.

chose over 150 different Martian sub-terrain samples. Training of the Hamming net learning convergence is achieved in less than two hundred iterations. Figure 5 demonstrates the results of our neural network terrain classifier for a number of different types of Martian terrains.

4.4 Fuzzy Logic-based Terrain Classifier

One good method of investing human knowledge in the terrain condition assessment process is by applying a fuzzy logic-based classifier. Human knowledge can be integrated in the form of a knowledge base (i.e., fuzzy rule sets and membership functions). A fuzzy logic classifier performs three distinctive functions. First, it performs fuzzification of terrain feature vectors that associate crisp input values to their corresponding fuzzy belief value via a group of non-linear membership functions. Secondly, it has a fuzzy logic controller that acts like an inference engine, and governs a set of fuzzy logic rules for associating range of applicability of fuzzy inputs to fuzzy outputs. Thirdly, it performs defuzzification of a fuzzy output value to an equivalent crisp value. Our Fuzzy Logic Terrain Classifier (FLTC) classifies the sub-terrain into five different terrain classes. Using the texture analysis method described in section 4.1, we initially compute image intensity contrast, energy, and variance, as well as rock blob size for each sub-

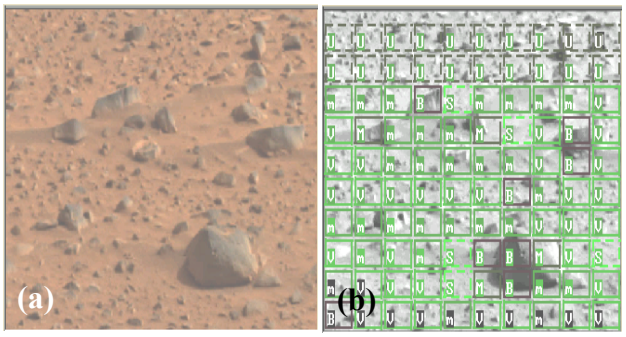


Figure 5. (a). Martian terrain; (b) Terrain assessment results of neural network terrain classifier. Classified sub-terrain designations: (S) sandy sub-terrain, (V) rough sub-terrain, (m) very rocky sub-terrain, (M) rocky sub-terrain, (B) impassable sub-terrain (i.e., big rocks formation), and (U) uncertain.

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terrain image window. The first three attributes yield the most consistent indication of the rock texture formation. The last attribute deals with physical size of the terrain salient features. The image texture attributes are used as input to the FLTC. Figure 6 and 7 present the input and output fuzzy membership functions of the FLTC respectively. The fuzzy inference engine of FLTC classifies the terrain condition based on a certainty confidence level that is indicative of degree of roughness of the terrain.

For development of the FLTC, we chose a set of natural terrain image samples randomly selected from among many salient features of different terrains. We compiled over 250 different patterns of natural sub-terrains and asked our human experts to classify terrain conditions per size of sub-terrains. In parallel, for each image sub-frame we computed the terrain surface texture attributes. By considering range of variation of terrain salient features texture attributes over entire set of sub-terrain samples, as well as the unique clues the human experts used to classify the sub-terrains, we have designed the fuzzy logic membership functions and fuzzy rules shown in the Figures 6 and 7. The FLTC closely mimics the human expert's judgment of the terrain traversability as in [3]. Figure 8 illustrates four different types of Martian terrains classified by the FLTC.

5. Technical Discussions

One important factor in terrain classification is *Terrain Observability*. Terrain observability refers to the realization of depth of view of terrain and the fact that terrain perception degrades with the distance of terrain salient features (i.e., rocks formations) from an observer's position. Many factors

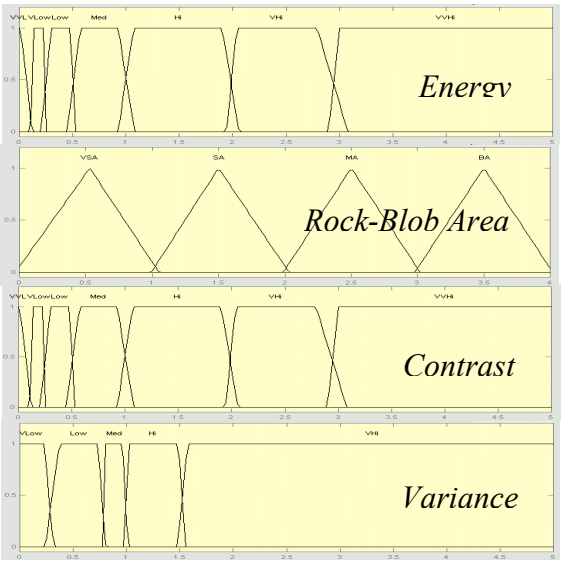


Figure 6. Input Membership Functions of Fuzzy Logic Rock Classifier

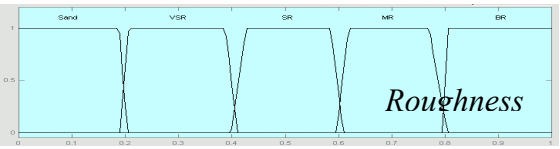


Figure 7. Output Membership Function of Fuzzy Logic Rock Classifier

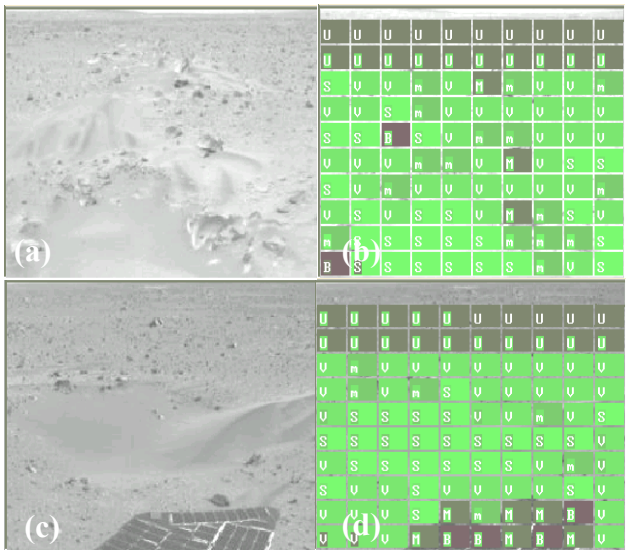


Figure 8. (a) and (c) are two different Martian Terrains; (b) and (d) represent terrain assessment results based on Fuzzy Logic Terrain Classifier. Classified sub-terrain designations are as defined in Figure 5.

contribute toward limiting terrain observability. The imperative factors are: environment conditions (i.e., lighting, shadows, humidity), digital camera resolution, sensitivity, and lens clarity and distortion, camera pose relative to ground plane, as well as many other peculiar terrain conditions, (i.e., upward or downward terrain slopes, trenches, hills, etc). In practice, it is, however, difficult to establish a concise mathematical model linking actual terrain features' attributes with the terrain image information content. To deal with this problem, we adjust terrain texture properties classified by the terrain classifiers using a piece-wise weight factor whose magnitude varies with the depth of view in the image. This piece-wise weight factor has unity magnitude and its order is gradually increased with the depth of view. The higher bound on this piece-wise weight factor is adaptively adjusted depending on the order of features found in the terrain after doing a primary blob analysis of it [8]. As you notice in Figures 5 and 8, both methods tend to classify the upper portion of terrain with high degree of uncertainty. This reflects poor terrain observability in the upper region of the image. Although this trait limits range of terrain traversability assessment of the robot, it is a more reliable and conservative approach for path planning in natural terrain [7]. For terrain features extraction, we initially enhance the terrain image and apply an adaptive binarization technique to isolate rock formations in the terrain. The adaptive binarization locally decides an optimum threshold level for binarization of an image sub-frame. The adaptive binarization is a relatively fast approach and yields a reasonable result. From the resultant binarized image, physical size, location, and pose of the rocks can be

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estimated via camera calibration techniques [19]. The orientation of first moment of rock blob area is a good indication of rock orientation. We use this technique to get a good sense of physical rock blob areas. By accurate digital imaging measurement technique, shape of each rock formation can be configured and modeled realistically and brought into a simulation environment to construct a virtual model of the terrain. The virtual terrain gives a better sense of perception of the terrain. Such a model can also be used for design and verification of trajectory path plans for a navigating mobile robot in natural terrain [8].

6. Comparative Evaluation

To evaluate performance of the three terrain classifiers for different classes of natural terrain, we conducted an experiment using a set of randomly selected Martian terrain images shown in Figure 9. The image test set contained images taken by the Mars Pathfinder Lander (1997) and *Spirit* Mars rover (2004) from different areas on Mars. The images were of size 320x240 pixels. We chose sub-terrain window size of 32x24 pixels for terrain assessment sampling that yields one hundred sub-terrains per terrain image. Figure 10 summarizes the performance results of the three classifiers.

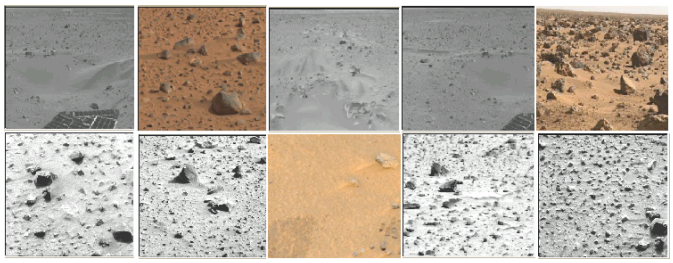


Figure 9. Ten different Martian terrains used for performance analysis of the terrain classifiers.

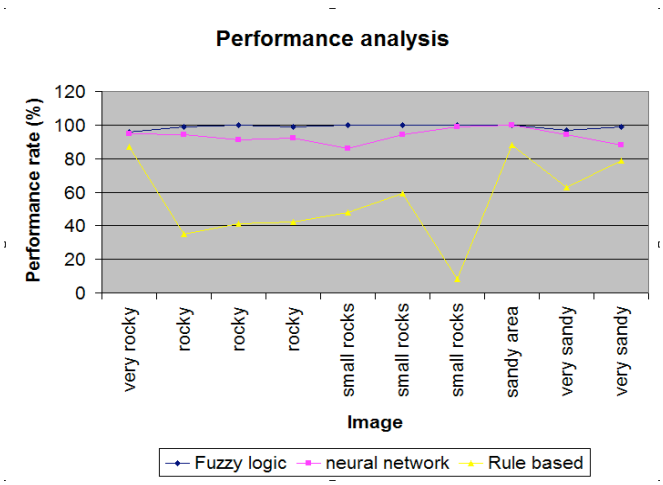


Figure 10. Terrain traversability assessment performance of three classifiers: (1) Rule-based Terrain Classifier, (2) Neural Network-based Terrain Classifier, (3) Fuzzy Logic-based Terrain Classifier.

The rule-based terrain classifier (RBTC) had the least performance characteristics of all three. The neural-network terrain classifier (NNTC) achieved very good performance with average success rate over 93 percent. However, the fuzzy logic-based terrain classifier (FLTC) achieved exceptional performance with average success rate over 98 percent. The RBTC is computationally fast. It applies only 10 rules for classification of all six states of the terrain conditions we examined. The NNTC was trained with over 144 terrain texture patterns. The proposed Hamming neural network terrain classifier can classify many terrain texture patterns. However, care must be taken not to over-specify or under-specify terrain texture pattern samples. The FLTC with 135 fuzzy rules is computationally not as fast as the other two classifiers but it is much more reliable for classifying the sub-terrain conditions accurately. All terrain analysis work presented herein, was performed using our Cooperative Robotic Man-Machine Interface (CORMI) application software running under FMCell robotic simulation and control software [20].

7. Conclusion

A mobile robot's adaptability to its environment is key for navigational endurance and longevity of the robot in natural terrain. In this paper, we have discussed some of the navigational challenges associated with path planning of mobile robots and presented three different methods including: rule-based, neural network-based, and fuzzy logic based for visual terrain mapping and traversability assessment of mobile robots in unstructured terrain environments. The proposed method applies visual imagery techniques for detection and localization of terrain salient features. The fuzzy logic and neural network-based terrain classifiers have shown promising results for a variety of the terrain conditions we examined. The proposed method has low computational overhead and hence is suitable for implementation onboard autonomous mobile robots, and perhaps future intelligent and autonomous Mars rovers.

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